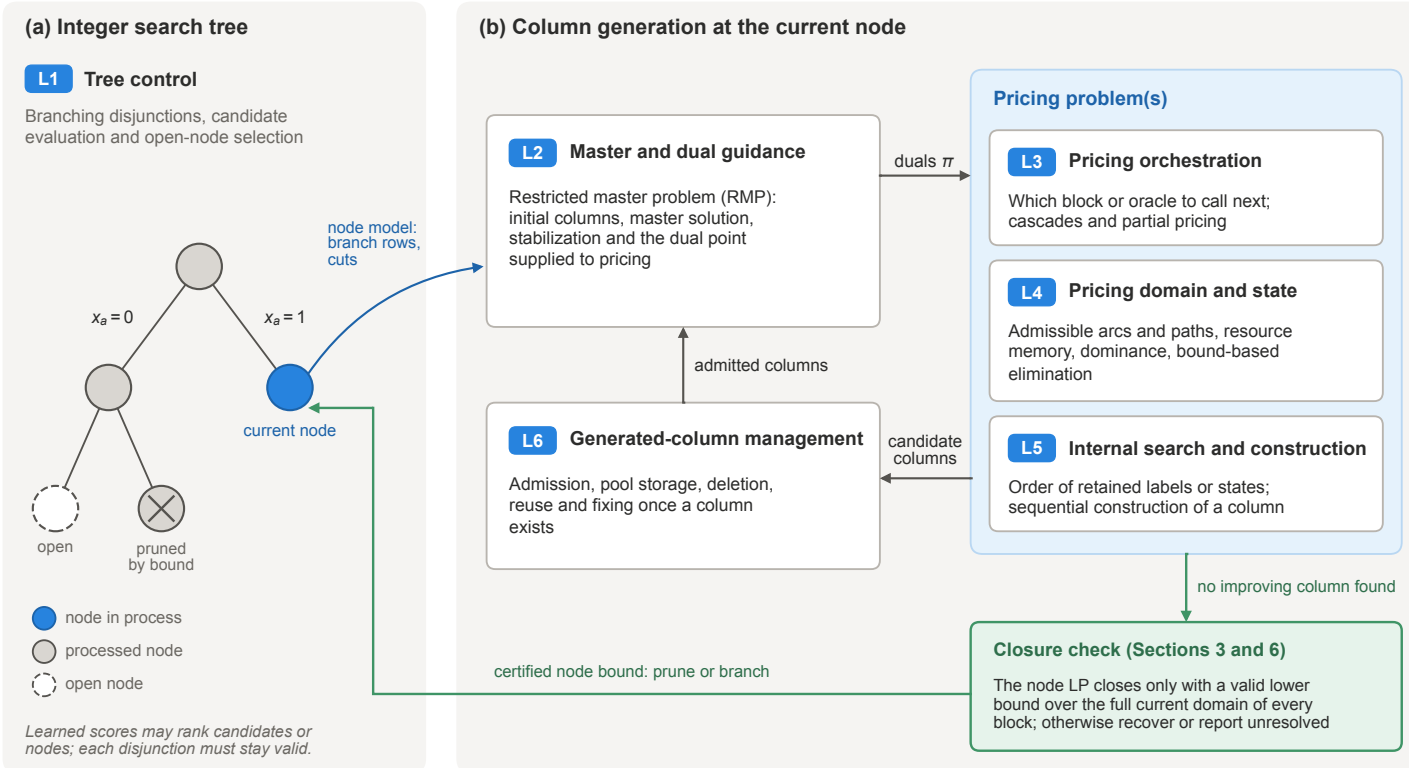

(a) Integer search tree
L1 Tree control
Branching disjunctions, candidate evaluation and open-node selection
node model: branch rows, cuts
$x_a = 0$
$x_a = 1$
current node
open
pruned by bound
node in process
processed node
open node
Learned scores may rank candidates or nodes; each disjunction must stay valid.
(b) Column generation at the current node
L2 Master and dual guidance
Restricted master problem (RMP): initial columns, master solution, stabilization and the dual point supplied to pricing
duals π
Pricing problem(s)
L3 Pricing orchestration
Which block or oracle to call next; cascades and partial pricing
L4 Pricing domain and state
Admissible arcs and paths, resource memory, dominance, bound-based elimination
L5 Internal search and construction
Order of retained labels or states; sequential construction of a column
candidate columns
admitted columns
L6 Generated-column management
Admission, pool storage, deletion, reuse and fixing once a column exists
no improving column found
Closure check (Sections 3 and 6)
The node LP closes only with a valid lower bound over the full current domain of every block; otherwise recover or report unresolved
certified node bound: prune or branch